From Measurements to Markov Graphs: Clustering-Based Framework for Dynamical Systems

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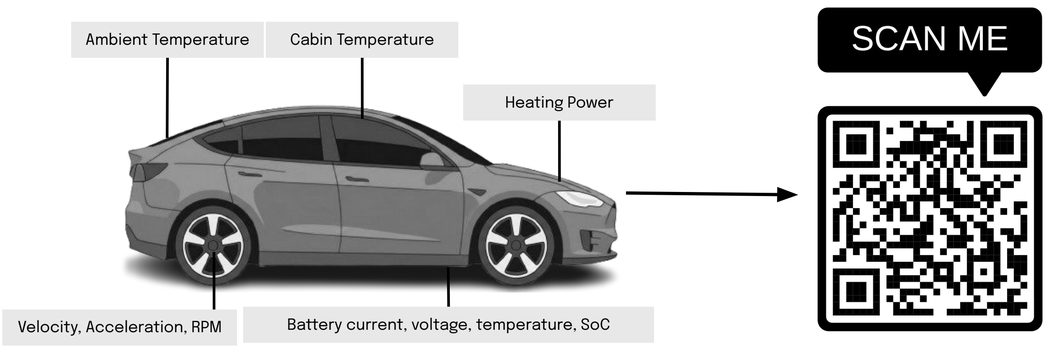


Fig 1. Illustration representing conversion of complex dynamical system into finite state transition dynamics. The QR code links to an example rendering of such state transition dynamics.

Characterizing real world systems is challenging

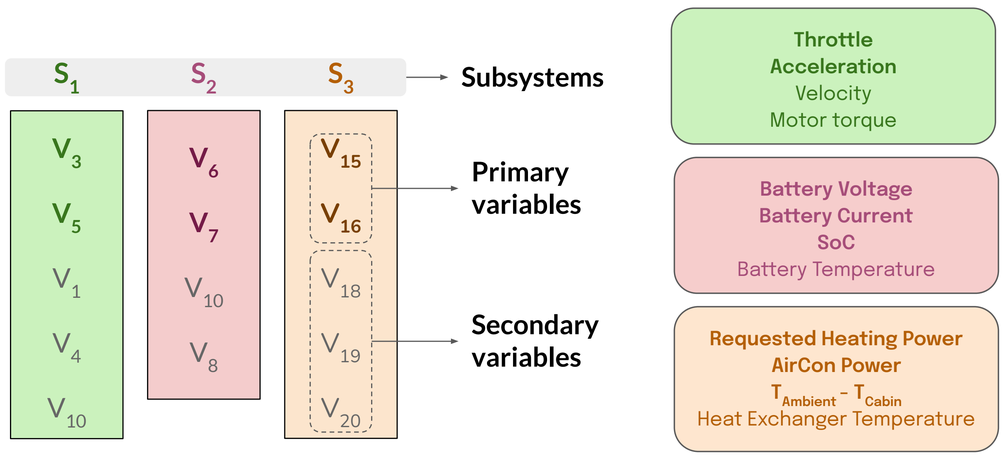
- Real world systems are **high-dimensional** (large number of variables)
- Large number of **evaluations** are required due to varied operating conditions
- Available datasets **[1]** are often **noisy** and **insufficient**, as experiments cannot span the full operating envelope

- How can we **characterize** system **performance** under different **operating conditions?**
- How can we reduce the **computational burden** of extensive system evaluations?
- How can we **systematically breakdown** and **analyse** such **high dimensional** system operating in **real world conditions**?

Classification of system variables into subsystems with finite set of variables

Objectives of subsystems classification

- Reduce the **number of variables** for analysis
- **Isolate** and **interpret** subsystem-specific dynamics.
- Identify **dominant behavioral regimes** within each subsystem



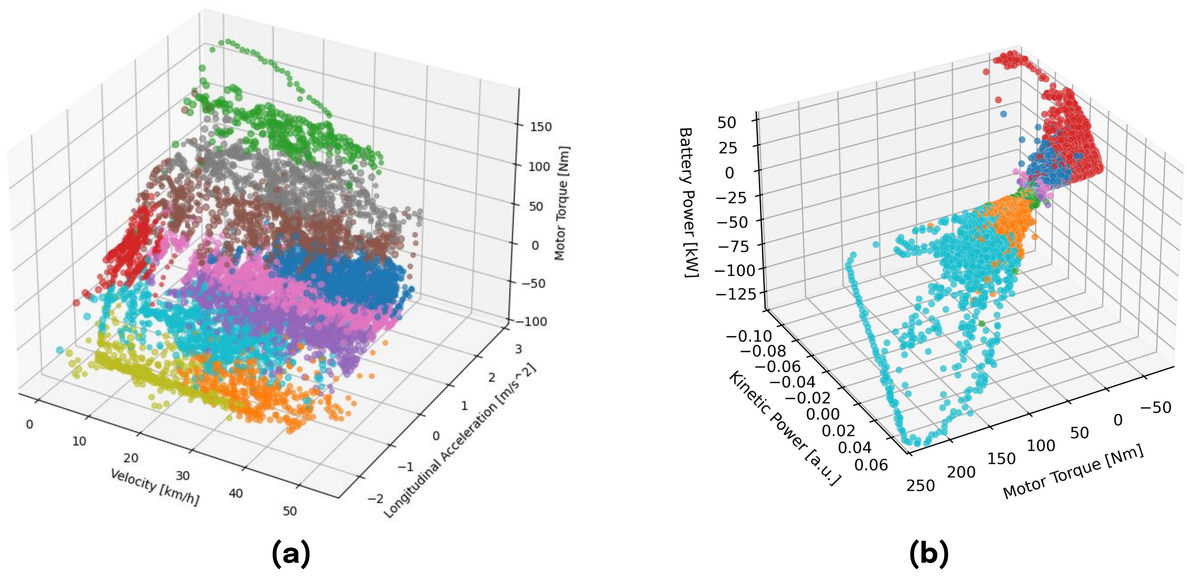


Fig 2. Clustering subsystem dynamics using K-means, (a) Representative drivetrain subsystem with 10 clusters, (b) Representative power subsystem with 10 clusters

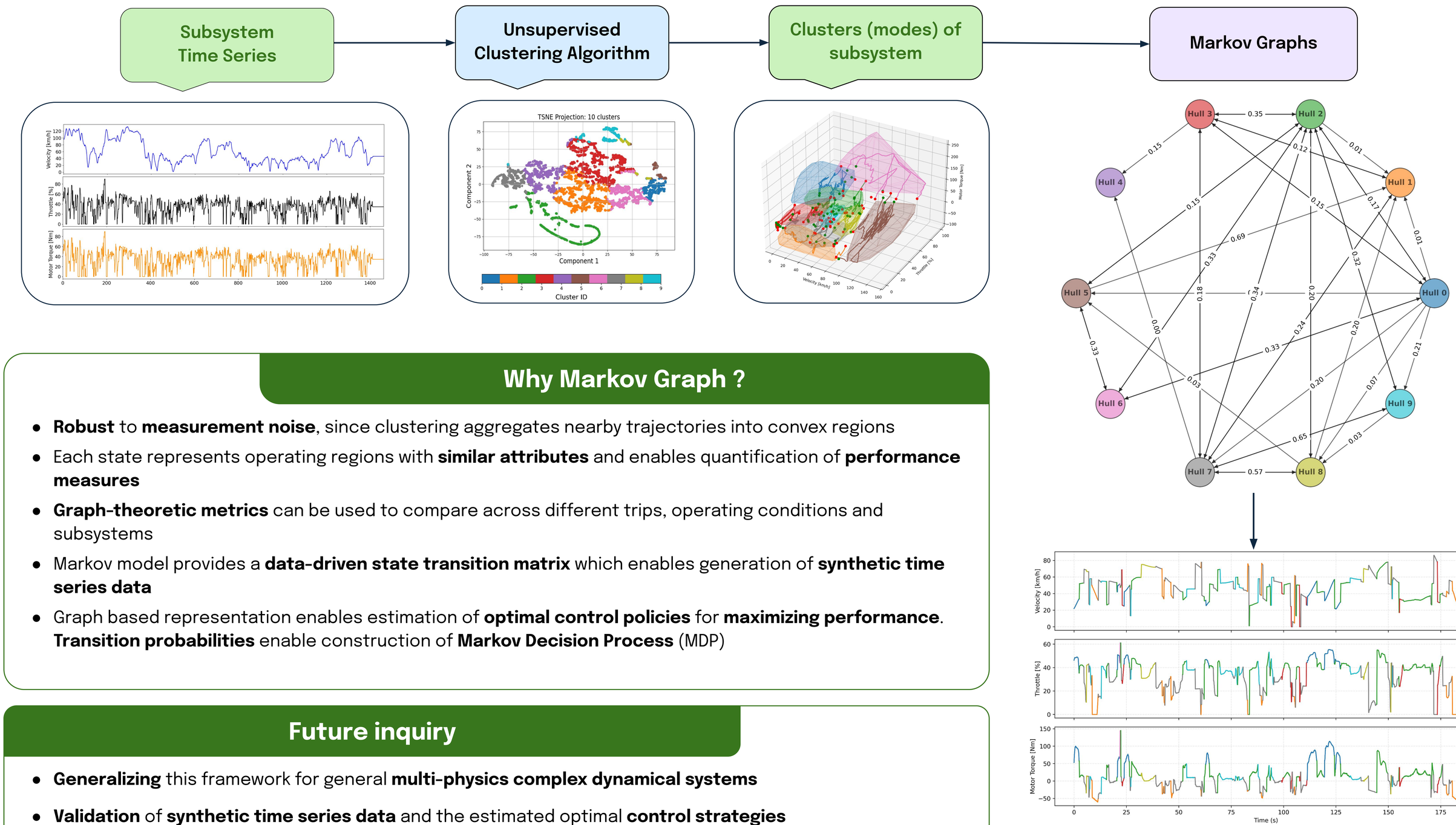


Fig 3. A realisation of synthetic time series data generated from markov graph for FTM (Fixed Testing Mode) with 100 hull transitions

Why Markov Graph ?

- **Robust** to **measurement noise**, since clustering aggregates nearby trajectories into convex regions
- Each state represents operating regions with **similar attributes** and enables quantification of **performance measures**
- **Graph-theoretic metrics** can be used to compare across different trips, operating conditions and subsystems
- Markov model provides a **data-driven state transition matrix** which enables generation of **synthetic time series data**
- Graph based representation enables estimation of **optimal control policies** for **maximizing performance**. **Transition probabilities** enable construction of **Markov Decision Process** (MDP)

Future inquiry

- **Generalizing** this framework for general **multi-physics complex dynamical systems**
- **Validation** of **synthetic time series data** and the estimated optimal **control strategies**
- Investigation of **subsystem coupling** via **conditional transition models**
- Exploration of **transfer learning** across vehicles or operating environments

References:

[1] Matthias Steinstraeter, Johannes Buberger, Dimitar Trifonov, 2020. Battery and Heating Data in Real Driving Cycles. Available at: https://dx.doi.org/10.21227/6jr9-5235.